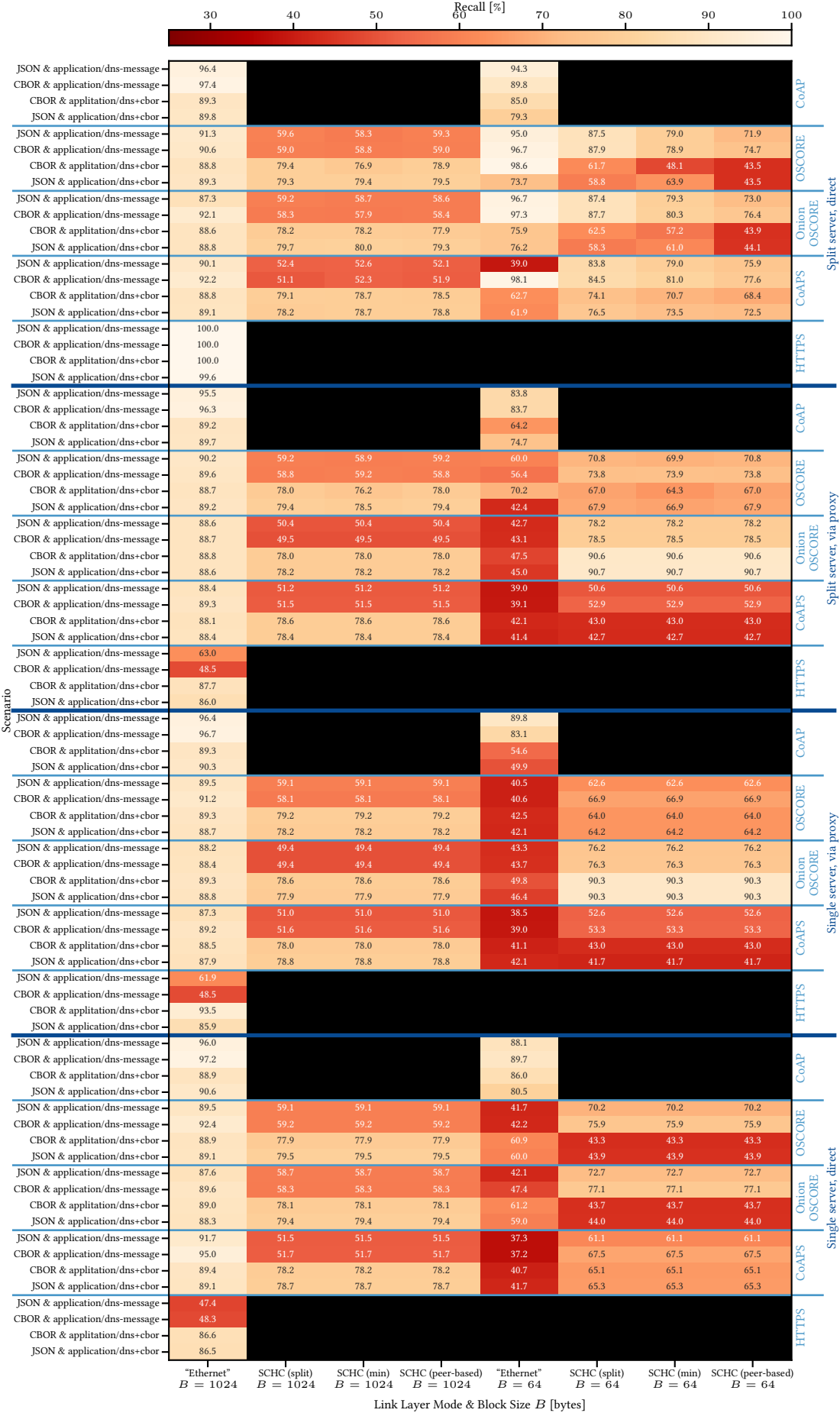
Recall [%]
30 40 50 60 70 80 90 100
Scenario
Link Layer Mode & Block Size B [bytes]
"Ethernet" $B = 1024$
SCHC (split) $B = 1024$
SCHC (min) $B = 1024$
SCHC (peer-based) $B = 1024$
"Ethernet" $B = 64$
SCHC (split) $B = 64$
SCHC (min) $B = 64$
SCHC (peer-based) $B = 64$
Split server, direct
JSON & application/dns-message 96.4 94.3
CBOR & application/dns-message 97.4 89.8
CBOR & applitation/dns+cbor 89.3 85.0
JSON & application/dns+cbor 89.8 79.3
CoAP
JSON & application/dns-message 91.3 59.6 58.3 59.3 95.0 87.5 79.0 71.9
CBOR & application/dns-message 90.6 59.0 58.8 59.0 96.7 87.9 78.9 74.7
CBOR & applitation/dns+cbor 88.8 79.4 76.9 78.9 98.6 61.7 48.1 43.5
JSON & application/dns+cbor 89.3 79.3 79.4 79.5 73.7 58.8 63.9 43.5
OSCORE
JSON & application/dns-message 87.3 59.2 58.7 58.6 96.7 87.4 79.3 73.0
CBOR & application/dns-message 92.1 58.3 57.9 58.4 97.3 87.7 80.3 76.4
CBOR & applitation/dns+cbor 88.6 78.2 78.2 77.9 75.9 62.5 57.2 43.9
JSON & application/dns+cbor 88.8 79.7 80.0 79.3 76.2 58.3 61.0 44.1
Onion OSCORE
JSON & application/dns-message 90.1 52.4 52.6 52.1 39.0 83.8 79.0 75.9
CBOR & application/dns-message 92.2 51.1 52.3 51.9 98.1 84.5 81.0 77.6
CBOR & applitation/dns+cbor 88.8 79.1 78.7 78.5 62.7 74.1 70.7 68.4
JSON & application/dns+cbor 89.1 78.2 78.7 78.8 61.9 76.5 73.5 72.5
CoAPS
JSON & application/dns-message 100.0
CBOR & application/dns-message 100.0
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 99.6
HTTPS
Split server, via proxy
JSON & application/dns-message 95.5 83.8
CBOR & application/dns-message 96.3 83.7
CBOR & applitation/dns+cbor 89.2 64.2
JSON & application/dns+cbor 89.7 74.7
CoAP
JSON & application/dns-message 90.2 59.2 58.9 59.2 60.0 70.8 69.9 70.8
CBOR & application/dns-message 89.6 58.8 59.2 58.8 56.4 73.8 73.9 73.8
CBOR & applitation/dns+cbor 88.7 78.0 76.2 78.0 70.2 67.0 64.3 67.0
JSON & application/dns+cbor 89.2 79.4 78.5 79.4 42.4 67.9 66.9 67.9
OSCORE
JSON & application/dns-message 88.6 50.4 50.4 50.4 42.7 78.2 78.2 78.2
CBOR & application/dns-message 88.7 49.5 49.5 49.5 43.1 78.5 78.5 78.5
CBOR & applitation/dns+cbor 88.8 78.0 78.0 78.0 47.5 90.6 90.6 90.6
JSON & application/dns+cbor 88.6 78.2 78.2 78.2 45.0 90.7 90.7 90.7
Onion OSCORE
JSON & application/dns-message 88.4 51.2 51.2 51.2 39.0 50.6 50.6 50.6
CBOR & application/dns-message 89.3 51.5 51.5 51.5 39.1 52.9 52.9 52.9
CBOR & applitation/dns+cbor 88.1 78.6 78.6 78.6 42.1 43.0 43.0 43.0
JSON & application/dns+cbor 88.4 78.4 78.4 78.4 41.4 42.7 42.7 42.7
CoAPS
JSON & application/dns-message 63.0
CBOR & application/dns-message 48.5
CBOR & applitation/dns+cbor 87.7
JSON & application/dns+cbor 86.0
HTTPS
Single server, via proxy
JSON & application/dns-message 96.4 89.8
CBOR & application/dns-message 96.7 83.1
CBOR & applitation/dns+cbor 89.3 54.6
JSON & application/dns+cbor 90.3 49.9
CoAP
JSON & application/dns-message 89.5 59.1 59.1 59.1 40.5 62.6 62.6 62.6
CBOR & application/dns-message 91.2 58.1 58.1 58.1 40.6 66.9 66.9 66.9
CBOR & applitation/dns+cbor 89.3 79.2 79.2 79.2 42.5 64.0 64.0 64.0
JSON & application/dns+cbor 88.7 78.2 78.2 78.2 42.1 64.2 64.2 64.2
OSCORE
JSON & application/dns-message 88.2 49.4 49.4 49.4 43.3 76.2 76.2 76.2
CBOR & application/dns-message 88.4 49.4 49.4 49.4 43.7 76.3 76.3 76.3
CBOR & applitation/dns+cbor 89.3 78.6 78.6 78.6 49.8 90.3 90.3 90.3
JSON & application/dns+cbor 88.8 77.9 77.9 77.9 46.4 90.3 90.3 90.3
Onion OSCORE
JSON & application/dns-message 87.3 51.0 51.0 51.0 38.5 52.6 52.6 52.6
CBOR & application/dns-message 89.2 51.6 51.6 51.6 39.0 53.3 53.3 53.3
CBOR & applitation/dns+cbor 88.5 78.0 78.0 78.0 41.1 43.0 43.0 43.0
JSON & application/dns+cbor 87.9 78.8 78.8 78.8 42.1 41.7 41.7 41.7
CoAPS
JSON & application/dns-message 61.9
CBOR & application/dns-message 48.5
CBOR & applitation/dns+cbor 93.5
JSON & application/dns+cbor 85.9
HTTPS
Single server, direct
JSON & application/dns-message 96.0 88.1
CBOR & application/dns-message 97.2 89.7
CBOR & applitation/dns+cbor 88.9 86.0
JSON & application/dns+cbor 90.6 80.5
CoAP
JSON & application/dns-message 89.5 59.1 59.1 59.1 41.7 70.2 70.2 70.2
CBOR & application/dns-message 92.4 59.2 59.2 59.2 42.2 75.9 75.9 75.9
CBOR & applitation/dns+cbor 88.9 77.9 77.9 77.9 60.9 43.3 43.3 43.3
JSON & application/dns+cbor 89.1 79.5 79.5 79.5 60.0 43.9 43.9 43.9
OSCORE
JSON & application/dns-message 87.6 58.7 58.7 58.7 42.1 72.7 72.7 72.7
CBOR & application/dns-message 89.6 58.3 58.3 58.3 47.4 77.1 77.1 77.1
CBOR & applitation/dns+cbor 89.0 78.1 78.1 78.1 61.2 43.7 43.7 43.7
JSON & application/dns+cbor 88.3 79.4 79.4 79.4 59.0 44.0 44.0 44.0
Onion OSCORE
JSON & application/dns-message 91.7 51.5 51.5 51.5 37.3 61.1 61.1 61.1
CBOR & application/dns-message 95.0 51.7 51.7 51.7 37.2 67.5 67.5 67.5
CBOR & applitation/dns+cbor 89.4 78.2 78.2 78.2 40.7 65.1 65.1 65.1
JSON & application/dns+cbor 89.1 78.7 78.7 78.7 41.7 65.3 65.3 65.3
CoAPS
JSON & application/dns-message 47.4
CBOR & application/dns-message 48.3
CBOR & applitation/dns+cbor 86.6
JSON & application/dns+cbor 86.5
HTTPS